# Sick Days and Schedule Type on Total Years of Service for T&E Workers

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# Introduction

**Objectives:**

- The main goal was to examine how the number of sick days and the type of schedule affect the total years of service of a train and engineer worker.
- The stakeholders are train and engineer workers as well as railroad companies

**Research Questions:**

- Research Question 1 – Does the number of sick days an employee takes significantly affect the total years of service?
- Research Question 2 – Does the type of schedule the employee has significantly affect the total years of service?

# Methods

**Data Collection:**

- Sampling Units: an individual train and engineer worker in the United States
- Final Sample Size: 160

**Measures:** This dataset was from Data.gov, which included the variables…

- Sick Days (days)
- Schedule (fixed or variable)
- Service (years)

**Analysis Method:**

- I used the software RStudio to run a generalized linear model for data analysis.

# Descriptives

Table 1 – Descriptive Statistics (n=160)

| | **Median** | **IQR** |
|---|---|---|
| **Service** | 13.917 years | 27.355 years |
| | **Median** | **IQR** |
| **Sick Days** | 3 days | 5.241 days |
| | **Fixed** | **Variable** |
| **Schedule** | 80 workers | 80 workers |

# Results

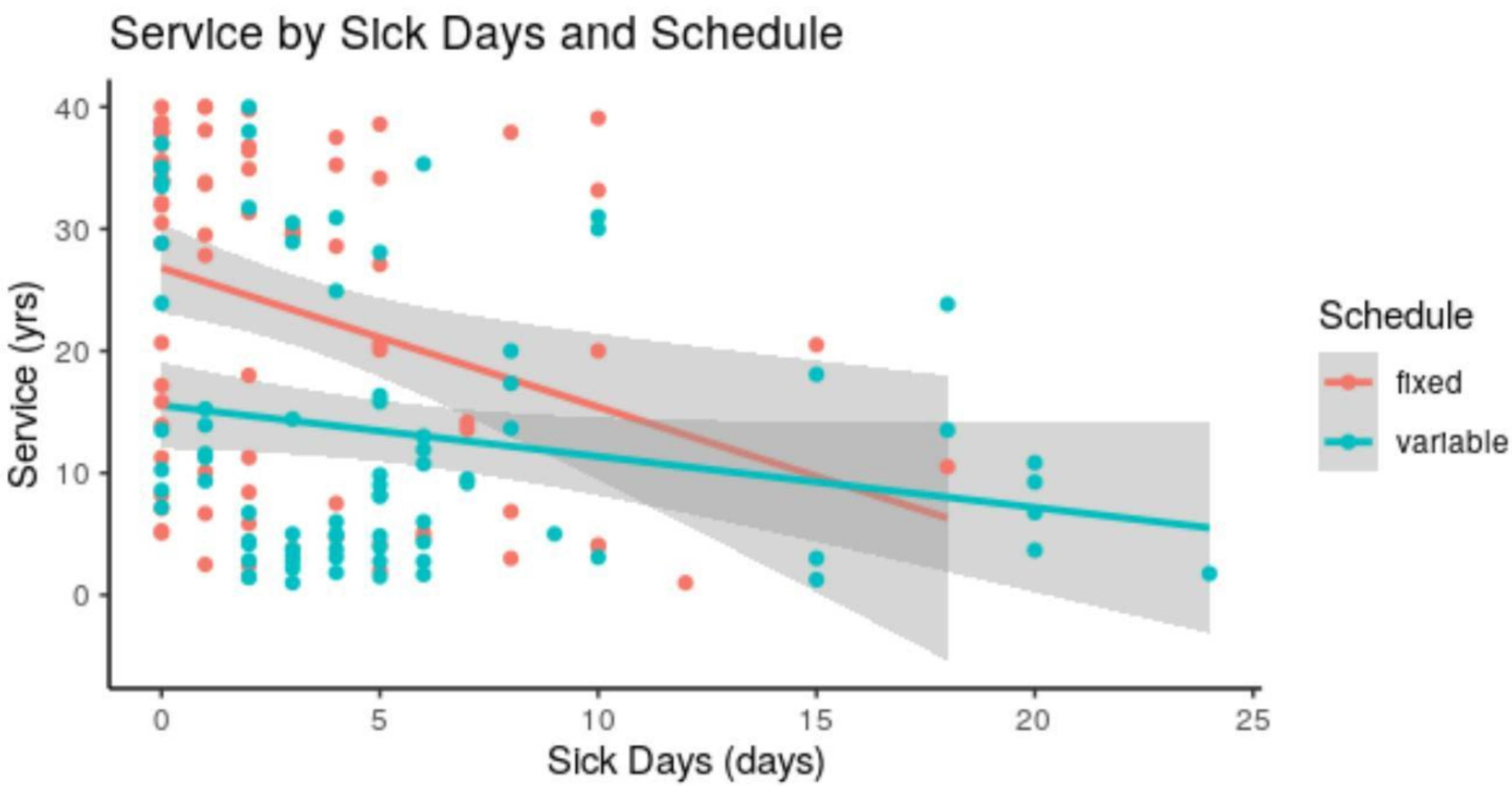
Table 2 – Model Results

| | Estimate | T-stat | St. Error | p-value |
|---|---|---|---|---|
| **Intercept** | 21.891 | 15.280 | 1.433 | < 2e-16 |
| **schedulevariable** | -8.161 | -4.084 | 1.999 | 7.06e-5 |
| **sick_days_c** | -1.139 | -3.199 | 0.356 | 0.00167 |
| **schedulevariable:sick_days_c** | 0.721 | 1.672 | 0.431 | 0.0966 |

**Model df:** 156

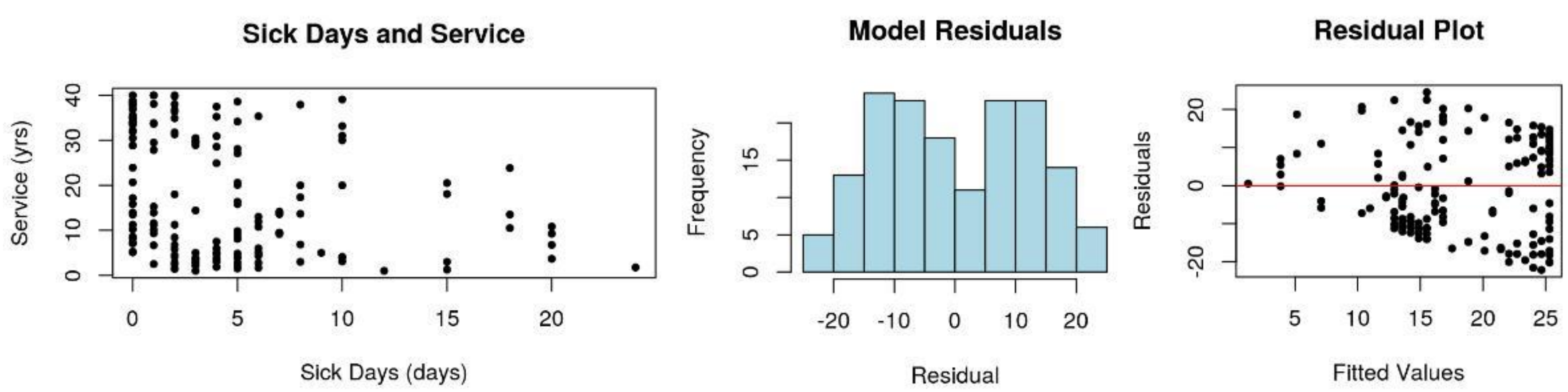
***Adjusted R-Squared***: 0.1935

## Interaction Plot:



# Assumptions

- The random sample assumption was not met because actively working BLET or UTU union members in the US were included in the sampling frame.
- The independent observation assumption was met because individual train and engineer workers were sampled.
- The linearity assumption was met because the scatterplot of sick days and service showed a negative linear relationship.
- The normality of the residual assumption was violated because the histogram of the residuals did not appear to have a normal distribution.
- The equal variance assumption was met because the residual plot showed a funneling pattern



# Discussion

**Interpretation:**

- Controlling the number of sick days, the type of schedule that a T&E worker has is a significant predictor of the total years of service ($t = -4.048$, $df = 156$, $p = 7.06e\text{-}5$). For T&E workers with an average number of sick days, having a variable schedule will decrease the total years of service by 8.161 years.
- Controlling for schedule, the number of sick days that a T&E worker takes is a significant predictor of the total years of service ($t = -3.199$, $df = 156$, $p = 1.67e\text{-}3$). For each increase of 1 sick day, the total years of service as a T&E worker with a fixed schedule decreases by 1.139 years.
- We found no significant interaction between the number of sick days and type of schedule on the total years of service of T&E workers ($t = 1.672$, $df = 156$, $p = 0.0966$).

**Confounding Variable:**

- Health insurance status is one confounding variable that may impact the results, in which those with health insurance would receive better healthcare that can prolong the service of T&E workers.

**Limitations:** Limitations include failing the random sample and normality assumption.

## Implications:

- We generalize that the number of sick days and the work schedule has a significant effect on the total years of service, which may lead to future policy proposals that can maximize the total years of service while also maintaining a good relationship between employees and their employers.
- For instance, if there seems to be a significant difference in the total years of service and work schedule, the companies may readjust the schedule to better maximize years of service; therefore, there may be more negotiations happening in the future between employees and employers.

## Future Research:

- One possible area that can be expanded for future research is to utilize a more diverse sample, in which all the unions in the US are included in the sampling frame instead of just BLET and UTU members.
- Because marriage can be associate with certain obligations that can affect the total years of service of a T&E worker, one thing that would be changed when conducting this study again is controlling marital status.

## References:


Askildsen, J. E., Bratberg, E., & Nilsen, O. A. (2005). Unemployment, labor force composition and sickness absence: a panel data study. Health economics, 14(11), 1087–1101. https://doi.org/10.1002/hec.994

Schneider, D., & Harknett, K. (2019). Consequences of routine work-schedule instability for worker health and well-being. American Sociological Review, 84(1), 82–114. https://doi.org/10.1177/0003122418823184

Work schedules and sleep patterns of railroad employees—Train and engine service (Railroad) data. (2018). [Data set]. Federal Railroad Administration. https://catalog.data.gov/dataset/work-schedules-and-sleep-patterns-of-railroad-employees-train-and-engine-service-railroad-